

SELBU, Norway

WMO: 012730

Lat: 63.225N

Lon: 11.008E

Elev: 525

StdP: 14.42

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	2.4	-7.1	3.8	-2.4	-2.1	5.0	3.9	27.7	29.7	23.3	31.6	2.3	150	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	78.2	62.3	74.0	60.2	70.3	58.6	64.0	74.7	61.8	71.2	59.8	68.5	6.4	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.3	77.0	69.2	57.4	71.6	66.5	55.5	66.9	63.8	29.5	75.3	27.8	71.4	26.5	69.0	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	16.9	14.0	DB	-4.5	83.5	7.4	3.9	-9.8	86.2	-14.2	88.5	-18.3	90.7	-23.7	93.5	(4)
(5)			WB	-4.6	67.4	6.9	2.9	-9.6	69.4	-13.6	71.1	-17.5	72.7	-22.5	74.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	25.0	26.8	32.0	39.0	47.3	53.1	59.2	57.2	51.3	41.2	32.5	27.3
	DBStd	14.01	10.16	9.79	7.42	5.65	7.01	5.67	6.28	4.90	5.14	5.99	8.57	9.86
	HDD50	4042	775	649	559	334	138	28	2	3	44	279	526	705
	HDD65	8762	1240	1069	1024	779	548	358	203	247	412	736	976	1170
	CDD50	783	0	0	0	4	55	121	288	225	82	8	0	0
	CDD65	29	0	0	0	0	0	2	23	4	0	0	0	0
	CDH74	373	0	0	0	0	15	48	266	43	1	0	0	0
	CDH80	79	0	0	0	0	1	5	71	2	0	0	0	0
Wind	WSAvg	5.1	6.2	5.6	6.0	4.9	4.9	4.3	4.1	4.3	4.7	5.5	5.4	5.5
Precipitation	PrecAvg	36.1	2.9	2.8	2.5	1.7	2.5	3.7	3.9	3.9	3.6	3.1	2.7	2.8
	PrecMax	44.4	6.8	4.7	5.4	5.2	3.9	6.3	6.5	6.4	8.9	6.5	8.0	5.9
	PrecMin	27.2	0.3	0.5	1.1	0.6	1.5	1.3	1.5	1.7	0.9	1.1	0.1	0.6
	PrecStd	4.7	1.7	1.2	1.3	1.1	0.7	1.1	1.5	1.5	2.1	1.3	2.2	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.4	46.5	49.3	65.2	75.9	80.0	87.0	78.5	71.4	61.9	52.7	46.5
		MCWB	38.4	39.3	39.4	50.2	56.9	61.2	66.7	62.2	57.4	51.0	46.6	40.9
	2%	DB	41.3	42.9	45.7	59.0	71.0	72.6	81.1	74.3	67.0	56.8	48.4	43.6
		MCWB	36.4	36.7	38.7	46.5	55.5	58.4	64.7	60.3	56.7	49.1	43.1	38.2
	5%	DB	39.3	40.7	43.4	53.6	66.1	68.0	76.0	70.8	63.6	53.7	45.7	41.3
		MCWB	35.0	35.5	37.6	43.5	53.3	56.8	62.1	59.2	54.7	47.1	40.7	36.4
	10%	DB	37.1	38.7	41.5	49.1	60.7	64.1	71.8	67.5	60.5	50.4	43.3	39.2
		MCWB	33.7	34.4	36.0	40.9	50.5	54.5	60.5	57.9	53.0	44.4	39.1	34.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.8	39.8	42.2	50.6	59.6	62.3	69.0	64.5	60.2	54.4	46.8	41.6
		MCDB	42.9	45.1	46.4	64.1	72.5	73.9	83.4	73.8	67.6	57.3	51.3	45.3
	2%	WB	37.3	37.7	40.0	46.9	56.5	59.8	66.2	62.1	57.6	49.7	43.9	39.1
		MCDB	40.3	41.5	44.4	57.3	68.5	71.3	79.0	71.0	65.0	56.1	47.5	42.6
	5%	WB	35.6	36.0	38.2	43.8	53.9	57.5	63.4	60.4	55.6	47.5	41.4	36.9
		MCDB	38.5	39.7	42.6	51.9	65.4	66.4	73.9	68.5	62.1	53.0	44.6	40.4
	10%	WB	34.0	34.7	36.6	41.6	51.0	55.4	61.2	58.5	53.7	45.2	39.3	35.3
		MCDB	36.6	38.2	40.7	48.3	60.1	62.7	70.4	65.9	59.7	49.7	43.2	38.3
Mean Daily Temperature Range	5% DB	MDBR	10.8	12.1	14.3	16.8	17.5	15.4	17.0	16.7	14.9	12.7	10.1	10.5
		MCDBR	10.5	11.3	14.5	24.4	27.4	23.8	26.0	22.7	20.5	16.8	12.7	11.6
		MCWBR	7.5	7.7	9.2	13.6	14.3	12.3	12.4	11.6	11.9	10.8	8.9	8.8
	5% WB	MCDBR	9.4	10.8	12.8	21.7	26.1	21.5	24.1	19.7	19.3	15.1	12.4	11.1
		MCWBR	7.1	8.0	8.6	12.5	14.0	11.9	12.3	11.0	11.7	10.1	9.3	8.7
Clear-Sky Solar Irradiance	taub	0.217	0.277	0.309	0.338	0.346	0.345	0.355	0.343	0.318	0.298	0.230	0.356	
	taud	2.033	2.213	2.235	2.302	2.414	2.466	2.458	2.487	2.541	2.449	2.114	2.701	
	Ebn at Noon	130	210	249	267	276	278	271	265	251	207	124	76	
	Edh at Noon	10	19	28	32	32	31	30	27	21	16	9	6	
All-Sky Solar Radiation	RadAvg	56	233	649	1222	1475	1520	1465	1126	684	303	83	27	
	RadStd	8	33	82	93	96	149	152	126	69	40	13	2	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-256	-320	N/A	N/A

Nomenclature: See separate page